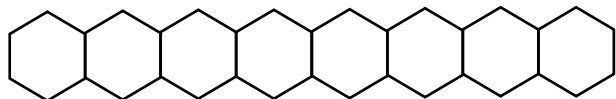


Series : ZYWX1

SET ~ 1

रोल नं.

Roll No.



कोड नं.
Q.P. Code

56/1/1

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-
पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code
on the title page of the answer-book.

नोट / NOTE

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 7 हैं।
Please check that this question paper contains 7 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 25 प्रश्न हैं।
Please check that this question paper contains 25 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की एयर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख्य-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



CHEMISTRY

Solutions



निर्धारित समय: 2 घंटे

Time Allowed : 2 hours

अधिकतम अंक : 50

Maximum Marks : 50



General Instructions:

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **25** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **FIVE** sections- Sections **A, B, C, D** and **E**.
- (iii) In **Section A** : Question numbers **1** to **13** are Multiple Choice (MCQ) type questions. Each question carries **1** mark.
- (iv) In **Section B**: Question numbers **14** to **17** are Very Short Answer (VSA) type questions. Each question carries **2** marks.
- (v) In **Section C**: Question numbers **18** to **22** are Short Answer (SA) type questions. Each question carries **3** marks.
- (vi) In **Section D** : Question number **23** is a Case Study-Based question carrying **4** marks.
- (vii) In **Section E** : Question numbers **24** and **25** are Long Answer (LA) type questions. Each question carries **5** marks.
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.
- (ix) Kindly note that there is a separate question paper for Visually Impaired candidates.
- (x) Use of calculators is **NOT** allowed.

SECTION - A

*This section contains **13** Multiple Choice Questions carrying **1** mark each.*

1. A compound $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ undergoes complete dissociation in water. The Van't Hoff factor 'i' is **1**
 - (A) 9
 - (B) 6
 - (C) 3
 - (D) 4
2. The value of Henry's constant K_H **1**
 - (A) Increases with decrease in temperature
 - (B) Decreases with increase in temperature
 - (C) Increases with increase in temperature
 - (D) Remains constant
3. An azeotropic solution of two liquids has a boiling point higher than either of the two when, it **1**



- (A) Shows a negative deviation from Raoult's Law
 (B) Shows a positive deviation from Raoult's Law
 (C) Is saturated
 (D) Shows no deviation from Raoult's Law

4. When 1 mole of benzene is mixed with 1 mole of toluene. The vapours will contain (Given: Vapour pressure of benzene = 12.8 kPa and vapour pressure of toluene = 3.85 kPa). 1

- (A) Equal amount of benzene and toluene as it forms an ideal solution.
 (B) Unequal amount of benzene and toluene as it forms a non-ideal solution.
 (C) Higher percentage of benzene
 (D) Higher percentage of toluene

5. How much ethyl alcohol must be added to 1 kg of water so that the solution will freeze at -14°C ? (K for water = 1.86°C/mol) 1

- (A) 7.5 mol (B) 8.5 mol
 (C) 10.5 mol (D) 9.5 mol

6. The table below shows the K_H values for some gases at 293 K at the same pressure.

KH values (kbar)	144.97	69.16	76.48	34.86
Gas	Helium	Hydrogen	Nitrogen	Oxygen

1

In which of the following are the gases arranged in their descending order of solubility (from left to right)?

- (A) Helium > Nitrogen > Hydrogen > Oxygen
 (B) Hydrogen > Helium > Nitrogen > Oxygen
 (C) Nitrogen > Hydrogen > Oxygen > Helium
 (D) Oxygen > Hydrogen > Nitrogen > Helium

7. Which of the following aqueous solutions will have the highest boiling point? 1

- (A) 1.0 M KCl (B) 1.0 M K_2SO_4
 (C) 2.0 M KCl (D) 2.0 M K_2SO_4

8. Out of the following 1.0 M aqueous solutions, which one will show the largest freezing point depression? 1



- (A) NaCl (B) Na₂SO₄
(C) C₆H₁₂O₆ (D) Al₂(SO₄)₃

9. The concentration by volume of a solution 'Y' used as an antifreeze in cars is 'X'. 'X' and 'Y' are respectively 1
- (A) 35 %, ethylene glycol (B) 37%, ethylene glycol
(C) 36 %, chloroform (D) 40 %, ethanol
10. In which of the following cases, blood cells will shrink? 1
- (A) When placed in water containing more than 0.9% (mass/volume) NaCl solution.
(B) When placed in water containing less than 0.9% (mass/volume) NaCl solution.
(C) When placed in water containing 0.9% (mass/volume) NaCl solution.
(D) When placed in distilled water.
11. Assume three samples of juices A, B and C having glucose as the only sugar present in them. The concentration of sample A, B and C are 0.1 M, 0.5 M and 0.2 M respectively. Freezing point will be highest for the fruit juice: 1
- (A) A (B) B
(C) C (D) All have same freezing point.

For question numbers 12 and 13- two statements are given: one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions using the codes (A), (B), (C) and (D) as given below :

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.
12. Assertion (A) : Brass is a mixture of copper and zinc, whereas German silver is a mixture of copper and tin. 1
Reason (R) : 1 ppm of fluoride ions in water causes the tooth to become mottled.



13. *Assertion (A)* : $\Delta H > 0$ in case of non-ideal solutions and $\Delta H = 0$ in case of ideal solutions. 1
Reason (R) : $\Delta V > 0$ in case of ideal solutions and $\Delta V = 0$ in case of non-ideal solutions.

SECTION - B

This section contains 4 Very Short Answer (VSA) type questions carrying 2 marks each.

14. Give reasons: 2
- (a) Cooking is faster in pressure cooker than in an open pan.
- (b) On mixing liquid X and liquid Y, volume of the resulting solution decreases. What type of deviation from Raoult's law is shown by the resulting solution? What change in temperature would you observe after mixing liquids X and Y?
15. (a) State Henry's Law and explain why the tanks used by scuba divers are filled with air diluted with helium (11.7% helium, 56.2% nitrogen and 32.1% oxygen). 2
- (b) What is anoxia?
16. (a) Explain the process of reverse osmosis using the example of a water desalination plant. Draw a diagram to better illustrate your points. 2
- (b) What happens when wilted flowers are placed in fresh water?
17. Why is Raoult's law considered a special case of Henry's Law? 2

SECTION - C

This section contains 5 Short Answer (SA) type questions carrying 3 marks each.

18. Henry's law constant for CO_2 in water is 1.67×10^8 Pa at 298 K. Calculate the number of moles of CO_2 in 500 ml of soda water when packed under 2.53×10^5 Pa at the same temperature. 3
19. Benzene and toluene form ideal solutions over the entire range of composition. The vapour pressure of pure benzene and toluene at 300K are 50.71 mm Hg and 32.06 mm Hg respectively. Calculate the mole fraction of benzene in vapour phase if 80 g of benzene is mixed with 100 g of toluene. 3
20. (a) Out of 0.1 molal aqueous solution of glucose and 0.1 molal aqueous solution of KCl, which one will have higher boiling point and why? 3



- (b) Predict whether Van't Hoff factor is less than 1 or greater than 1 in the following:
- CH_3COOH dissolved in water.
 - CH_3COOH dissolved in benzene.
21. Calculate the molarity of 9.8% (w/w) solution of H_2SO_4 if the density of solution is 1.02 gm/mL. 3
22. Some ethylene glycol ($\text{HOCH}_2\text{CH}_2\text{OH}$) is added to your car cooling system along with 5 kg of water. If the freezing point of water-glycol solution is -15°C , what is the boiling point of the solution? ($K_b = 0.52\text{ K kg/mol}$, $K_f = 1.86\text{ K kg/mol}$) 3

SECTION - D

This section consists of 1 case-based/source-based questions carrying 4 marks each.

23. The membranes involved in the process of osmosis may be of animal or vegetable origin and these occur naturally such as pig's bladder or parchment or can be synthetic such as cellophane. These membranes appear to be continuous sheets or films, yet they contain a network of submicroscopic holes or pores. Small solvent molecules, like water, can pass through these holes but the passing of bigger molecules like solute is hindered. Membranes having this kind of properties are known as semipermeable membranes (SPM). 4
- What happens when people consume a lot of salt or salty food?
 - How is meat preserved in the food industry?
 - Boiling point of water at 750 mm Hg is 99.63°C . How much sucrose is to be added to 500 g of water such that it boils at 100°C ? ($K_b = 0.52\text{ K kg/mol}$)

SECTION - E

This section consists of 2 Long Answer (LA) type questions carrying 5 marks each.

24. (a) Write the conditions necessary for the formation of ideal solutions. 5
- (b) State whether the following solutions are ideal or non-ideal. If solution is non-ideal, state whether it will show positive or negative deviation from Raoult's Law.
- Bromoethane and chloroethane
 - Methanol and acetone
 - Carbon disulphide and acetone
 - Benzene and toluene
- (c) Calculate the osmotic pressure in pascals exerted by a solution prepared by dissolving 1.0 g of polymer of molar mass 185,000 in 450 mL of water at 37°C .



-
25. 0.6 mL of acetic acid (CH_3COOH) having density 1.06 g/mL, is dissolved in 1 litre of water. The depression in freezing point observed for this strength of acid was 0.0205°C . Calculate the van't Hoff factor and the dissociation constant of acid.

5

